

Regression Testing Pipeline

Daily automated safety evaluation of AI psychiatric companion conversations

1. CI/CD

GitHub Actions workflow: regression-test.yml

Triggers: daily schedule (06:00 UTC) and manual dispatch; matrix: 2 rounds × 2 chat models = 4 generation jobs

2. Input datasets

Standardized Safety Dataset

Park et al. (2024)

Open-ended Dataset

HuggingFace everyday-conversations-llama3.2-1k

3. Conversation generation

Simulated multi-turn dialogue; output stored as dataset-round- $\{2,3\}$ - $\{\text{model}\}$.json with Langfuse tracing

Companion agent

GPT-4o / GPT-5

In-silico processor

Up to 5 turns / session

Psychiatrist agent

GPT-4o / GPT-5

4. Automated safety evaluation

LlamaGuard

Llama-Guard-4-12B

- 14 safety categories (S1–S14)
- Per-message classification

OpenAI moderation

omni-moderation-2024-09-26

- 21 moderation categories
- Per-message scores

LLM-as-judge

OpenAI Batch API

- Structured rating rubric
- Batch processing (JSONL)

5. Result storage

GitHub artifacts

Per-run snapshots; 7-day retention

Google Drive (shared drive)

Date-partitioned long-term archive

6. Post-processing and analysis

Safety trends

agg_reg_test.py

Moderation agreement

reg_test_moderation_agreement.py

Category breakdown

reg_test_categories_plots.py

Diversity metrics

self_bleu.py /
unique_n_grams.py

LLM judge ratings

reg_test_llm_judge_ratings_boxplot.py

LaTeX tables

reg_test_table.py